

URINARY SYMPTOMS

Give urgent attention to the patient with urinary symptoms and any of:

- Unable to pass urine with lower abdominal discomfort/distention
- Blood/protein in urine and new swelling of face/feet, BP $\geq$ 140/90 or passing little urine: **kidney disease** likely
- Blood in urine and sudden, severe, one-sided pain in flank or groin: **kidney stone** likely

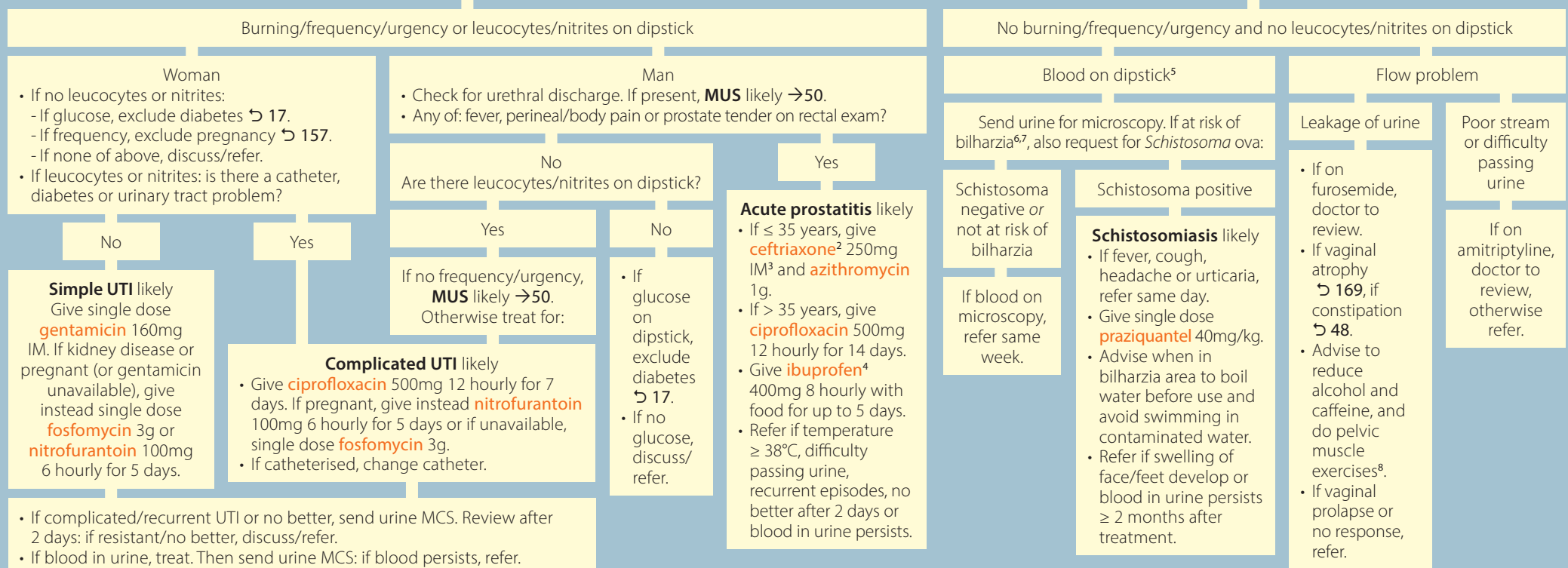
Manage and refer urgently:

- If unable to pass urine, insert urinary catheter.
- If **kidney disease** likely: if pulse $>$ 100 or respiratory rate $\geq$ 30, give face mask oxygen and **furosemide** 80mg slow IV, avoid IV fluids. If BP $>$ 150/100, give **amlodipine** 5mg and **furosemide** 40mg orally.
- If **kidney stone** likely: give **sodium chloride 0.9%** 1L IV 6 hourly. If pain severe, give **morphine** 10mg IM or 3-10mg slow IV. For IV: dilute 10mg morphine with 9mL of sodium chloride 0.9%.
- If **complicated pyelonephritis** likely: first collect urine for MCS and then give **ceftriaxone** 1g IV¹/IM. If BP $<$ 90/60, give **sodium chloride 0.9%** 500mL IV over 30 minutes, repeat until systolic BP $>$ 90. Continue 1L 6 hourly. Stop if breathing worsens.

- Flank pain with leucocytes/nitrites on urine dipstick, and any of: vomiting, BP $<$ 90/60, pulse $\geq$ 100, diabetes, male, pregnant or post menopause: **complicated pyelonephritis** likely

Approach to the patient with urinary symptoms not needing urgent attention

If flank pain with leucocytes/nitrites, **uncomplicated pyelonephritis** likely: send urine MCS. Give **ciprofloxacin** 500mg 12 hourly for 7 days and **paracetamol** 1g 4-6 hourly. Advise to return if worse: refer.



¹Do not mix Ringer's lactate and IV ceftriaxone. Flush IV line with **sodium chloride 0.9%** before and after IV ceftriaxone. ²If severe penicillin allergy (history of anaphylaxis, urticaria or angioedema), omit ceftriaxone and increase **azithromycin** to 2g.

³For ceftriaxone 250mg IM injection: dissolve 250mg in 0.9mL **lidocaine 1%** without epinephrine (adrenaline). ⁴Avoid if peptic ulcer, asthma, hypertension, heart failure or kidney disease. ⁵If menstruating, repeat dipstick after period has finished.

⁶Patient at risk of bilharzia if s/he has washed/swam in dams, streams or lakes in an endemic area (Limpopo, North West, Mpumalanga, KwaZulu-Natal and parts of Eastern Cape). ⁷If microscopy not available and patient lives in endemic area, treat as schistosomiasis. ⁸Repeated contraction and relaxation of pelvic floor muscles.